THE ETA TRANSFORM REDUCED MODULO THE ELEMENT IS QUASI-ISOMORPHIC TO THE BOCKSTEIN COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The counted statement span refers to the Bockstein complex by its numbered equation label, delivered verbatim. The author notes inside the counted proof span that all tensor products are over A; that sentence is reproduced. The two references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- The eta transform eta_f M^bullet of a complex of f-torsion free modules and its grading. used at proof lines 27753-27754. Supplies the source complex and the description of its degree i part as f^i M^i.
- The Bockstein complex H^bullet(M^bullet/f) with its differentials beta. named in the statement at line 27748 and used at proof lines 27768-27769. Is the target of the canonical map and the source of the compatibility computation.
- Vanishing of the kernel of the comparison maps (eta_f M)^i tensor A/fA -> Z^i/B^i and acyclicity. invoked at proof lines 27774-27779. Is what upgrades the map of complexes to a quasi-isomorphism.

Source reference. Original equation 27686: sources/stacks-more-algebra/pinned-original-source.tex, source lines 27686–27686. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original remark 27579: sources/stacks-more-algebra/pinned-original-source.tex, source lines 27579–27579. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let A be a ring and let $f \in A$ be a nonzerodivisor. Let $M^\bullet$ be a complex of f-torsion free A-modules. There is a canonical map of complexes*

$$\eta_f M^\bullet \otimes_A A/fA \longrightarrow H^\bullet(M^\bullet/f)$$

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

which is a quasi-isomorphism where the right hand side is the complex (27686).

Proof. Let $x \in (\eta_f M)^i$. Then $x = f^i x' \in f^i M$ and $d^i(x) = f^{i+1} y \in f^{i+1} M^{i+1}$. Thus d^i maps $x' \otimes f^i$ to zero in $M^{i+1} \otimes f^i A/f^{i+1}A$. All tensor products are over A in this proof. Hence we may map x to the class of $x' \otimes f^i$ in $H^i(M^\bullet \otimes f^i A/f^{i+1}A)$. It is clear that this rule defines a map

$$(\eta_f M)^i \otimes A/fA \longrightarrow H^i(M^\bullet \otimes f^i A/f^{i+1}A)$$

of A/fA-modules. Observe that in the situation above, we may view $x' \otimes f^i$ as an element of $M^i \otimes f^i A/f^{i+2}A$ with differential $d^i(x' \otimes f^i) = y \otimes f^{i+1}$. By the construction of β above we find that $\beta(x' \otimes f^i) = y \otimes f^{i+1}$ and we conclude that our maps are compatible with differentials, i.e., we have a map of complexes.

To finish the proof, we observe that the construction given in the previous paragraph agrees with the maps $(\eta_f M)^i \otimes A/fA \to Z^i/B^i$ discussed in Remark 27579. Since we have seen that the kernel of these maps is an acyclic subcomplex of $\eta_f M^\bullet \otimes A/fA$, the lemma is proved. $\square$